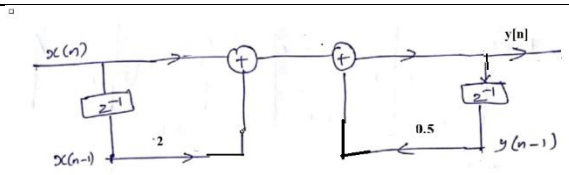
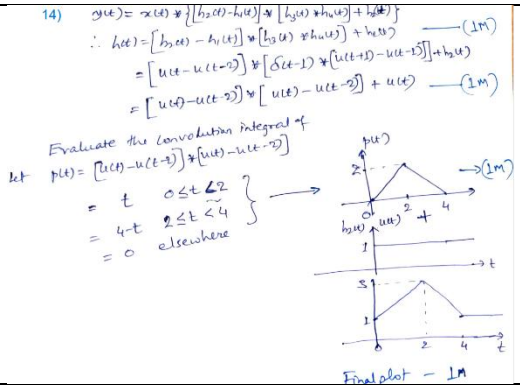
ECE 2123 SIGNALS AND SYSTEMS MID TERM SCHEME JULY-DEC 2025			TOT MARKS:30			
Qno	Faculty	**Scheme**(1-5 ARE MCQS)	Marks	LP	CLO	Blooms level
1	AV	C	1	1-2	1	3
2	AV	D	1	4	1	4
3	PMP	C	1	8	2	5
4	PMP	A	1	5,3	3	4
5	AV	B	1	10	4	3
6	DB	**(a) Causality: Yes** A system is causal if h[n] = 0 for all n < 0. Here, u[n] = 0 for n < 0, so h[n] = 0 for n < 0. Therefore, the system is CAUSAL. ------ 1 marks **(b) BIBO Stability: Yes** For an LTI system, BIBO stability requires: $\sum_{k=-\infty}^{\infty}\|h[n]\| < \infty$ $h[n] = \left(\frac{1}{2}\right)^n u[n] = \begin{cases} (1/2)^n, & n \geq 0, \\ 0, & n < 0. \end{cases}$ $\sum_{n=-\infty}^{\infty} \|h[n]\| = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n.$ $\sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = \frac{1}{1-\frac{1}{2}} = 2.$ Therefore, the system is BIBO-STABLE ---1 marks	2	15	4	4
7	AV		2	16	2	4
8	AV	X[n]={1,2} n=1,2 H[n]={1,1} n=0,1 Y[n]={1,3,2}----1m n={1,2,3}...1m	2	13	4	3

9	UV	8 ① $y(t) = t^2 x(t)$ $y(t)\|_{x(t)=x(t-T)}$ = o/p for delayed I/p $x(t-T)$ $= t^2 x(t-T)$ —(A) $y(t-T) = y(t)\|_{t=t-T}$ → o/p delayed by T units $= (t-T)^2 x(t-T)$ —(B) ∴ (A) ≠ (B) ⇒ not time invariant. —1 ② $y(t)\|_{x(t)=x(t-T)} = e^{2x(t-T)}$ $y(t-T) = e^{2x(t-T)}$ —1 ∴ time - invariant. ③ $y(t)\|_{x(t)=x(t-T)} = x(t-T)\sin 10\pi t$ $y(t-T) = x(t-T)\sin 10\pi(t-T)$ — 1 ∴ not time invariant.	3	9, 14	2	5
9	UV	(9) $x(t)$ —1 $x_e(t) = \frac{1}{2}(x(t)+x(-t))$ —1 $x_o(t) = \frac{1}{2}(x(t)-x(-t))$ —1	3	3, 5	3	3
10	AKT	Q(10) $x[n] = 2^n\{u[n+1]-u[n-3]\}$, Given. $x[n]$, $x[n+3]$, $x[n-3]$ — 1M each	3	7	3	2
11	AKT	Q.11) $x(t) = \begin{cases} 2-t, & 0 \le t \le 2 \\ 0, & \text{otherwise} \end{cases}$, Given. $x(t)$, $x(t-1)$, $x(0.5t-1)$ — 1M each	3	4, 6	1	4 OR 3
13	DB	Solution: $y_1[n] = \sum_{k=-\infty}^{n} x[k]$ **Invertible : Yes.** Inverse system: $x[n] = y_1[n] - y_1[n-1] \Rightarrow h_{inv}[n] = \delta[n] - \delta[n-1]$ **Time-invariant. Yes.** A shift in x produces the same shift in y_1.	3	11	4	3

		BIBO-stable? **No.** Bounded input x[n] = u[n] yields $y_1[n] = n+1$, which is unbounded. ----- each 0.5 Marks $y_2[n] = x[n^2]$ **Invertible : No.** The output only samples x[n] at perfect squares {0, 1, 4, 9, ...}, so values like x[2], x[3], ... are lost and cannot be recovered. **Time-invariant**. **No**. For a shift n_0: T{x[n-n_0]} = x[n^2 - n_0] while y_2[n-n_0] = x[(n-n_0)^2]; **BIBO-stable**? **Yes.** If |x[n]| ≤ B then |y_2[n]| = |x[n^2]| ≤ B for all n. ----- each 0.5 Marks				
14	PMP		4	8, 12	2	3